Lumiq

AI Product Feedback Agent

From messy user feedback to sprint-ready insights — in minutes

0 to 1 Product Build • Shijin • Product Manager

THE PROBLEM

The Feedback Problem

Product teams collect **hundreds of feedback rows every week** from reviews, support tickets, surveys, and user calls.

But most of it never becomes a decision.

- Hours spent manually reading feedback rows to find patterns
- Important signals buried under noise, emotions, and duplicates
- Decisions made on incomplete or misleading information

"App keeps crashing"
"भुगतान विफल"
"Love the UI!"
"yaar checkout toot gaya"
"Refund not processed"
"Slow loading"
"Can't cancel order"
"Great support!"
"OTP nahi aaya"
"Wrong item delivered"
"search broken"
"Price mismatch!"

This is what 200 feedback rows looks like without structure

PMs spend up to 40% of their time on feedback analysis that could be automated

Lumiq

THE INSIGHT

What if a PM could get structured, prioritized insights from 200 feedback rows in under 5 minutes?

No manual tagging

Agent classifies intent automatically — bug, complaint, feature request, churn, or praise

No subjective analysis

Root causes backed by evidence — not gut feel or whoever speaks loudest in the meeting

No guesswork on priority

Scored by impact, frequency, severity and confidence — sprint-ready every time

"The goal is not to summarize feedback. The goal is to help teams move from collection to confident decision-making."

THE SOLUTION

Meet Lumiq

An AI agent that thinks like a senior product analyst

Ingest messy feedback from any source. Get structured, prioritized insights — in one pipeline.

- Ingests CSV or Google Sheets
- Clusters, analyzes and prioritizes automatically
- Conversational agent for follow-up questions

Lumiq Dashboard — Analysis Complete

142	7	4m 52s	$0.09
Rows Analyzed	Clusters Found	Run Time	Cost

Top Insights

CRITICAL	Payment Processing Failures	14% • 98 score
HIGH	Checkout Abandonment	11% • 81 score
MED	Delivery Tracking Issues	9% • 63 score

100+ rows analyzed | **9** pipeline steps | Under **5 minutes**

ARCHITECTURE

The 9-Step Autonomous Pipeline

1	2	3	4	5	6	7	8	
INPUT	**INGEST**	**DETECT**	**TRANSLATE**	**CLASSIFY**	**CLEAN**	**CLUSTER**	**ROOT CAUSE**	**INSIGHTS**
CSV / Sheets	Load rows	Language	→ English	Intent tags	Sentiment	AI grouping	Claude Sonnet	Priority ranked

The agent runs all 9 steps automatically after a single trigger. No human in the loop.

CAPABILITIES

What Lumiq Does

Multilingual Support

Handles English, Hindi and Hinglish feedback automatically. Translates before analysis.

Intent Classification

Tags each row as bug, complaint, feature request, churn signal or praise.

Smart Clustering

Groups similar feedback into meaningful problem themes using TF-IDF and K-Means.

Root Cause Analysis

Claude Sonnet identifies *why* each problem exists — not just what users are saying.

Priority Scoring

Weighs each insight by impact, frequency, severity and confidence score.

Conversational Agent

Ask questions about your insights in plain English after every analysis.

CONVERSATIONAL AI

Ask Your Data Anything

Lumiq Agent — Powered by LangChain + Claude

What are the top 3 problems?

Payment failures ranked **first** — critical severity, **14% frequency**. Checkout #2 (11%), delivery tracking #3 (9%).

Why is payment ranked first?

Root cause: **inadequate timeout handling** and retry logic in the payment gateway integration.

User asked

"What are the top 3 problems?"

Payment failures ranked first — critical severity, 14% frequency...

User asked

"Why is payment ranked first?"

Root cause: inadequate timeout handling and retry logic in the payment gateway...

User asked

"What should engineering fix first?"

Priority #1: Payment Processing Failures — HIGH effort, CRITICAL severity, 98 impact score

Powered by LangChain agent with 5 specialized tools

TECHNICAL

System Architecture

Intelligence Layer

- **Claude Haiku**
  Classification
- **Claude Sonnet**
  Reasoning & Root Cause
- **LangChain**
  Agent Orchestration
- **TF-IDF + SVD**
  Embeddings

Backend

- **Python**
  Core Pipeline
- **FastAPI**
  REST API
- **SQLAlchemy**
  ORM
- **Supabase**
  PostgreSQL

Frontend

- **Next.js 16**
  Dashboard
- **Tailwind CSS**
  Styling
- **Recharts**
  Data Visualization
- **Vercel**
  Deployment

Every module built and validated independently before integration

ENGINEERING · PRODUCT

5 Decisions That Shaped the Product

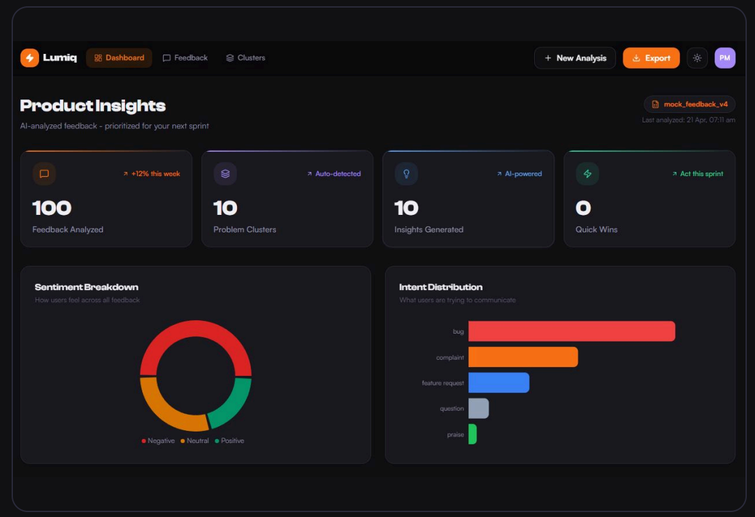
Every technical decision was a product decision in disguise

01	**K-Means over HDBSCAN** Algorithm selection for clustering	HDBSCAN failed with small deduplicated datasets. K-Means with Silhouette Score gave consistent clusters every time.
02	**Haiku for classify, Sonnet for reasoning** Model selection by task complexity	Match model capability to task. Saved **50×** on API costs without quality loss.
03	**Batch processing over single calls** API call optimization	100 rows in **6 seconds** vs 78 seconds. 92% reduction in latency and cost.
04	**TF-IDF over sentence-transformers** Embedding approach	90MB model crashed Render free tier. TF-IDF runs under **10MB** with equivalent quality.
05	**Sequential over parallel Claude calls** Concurrency model	Parallel threads competed for 512MB RAM. **Stable beats fast** when fast means broken.

Lumiq

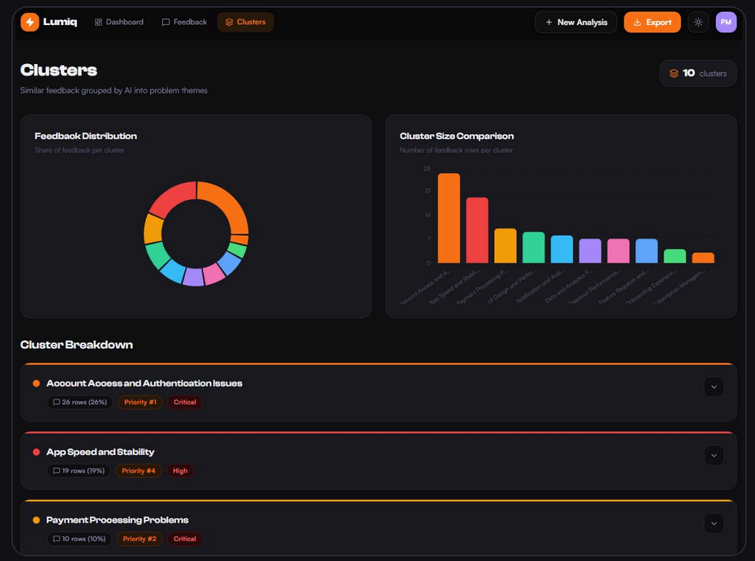
DEMO

Product Walkthrough



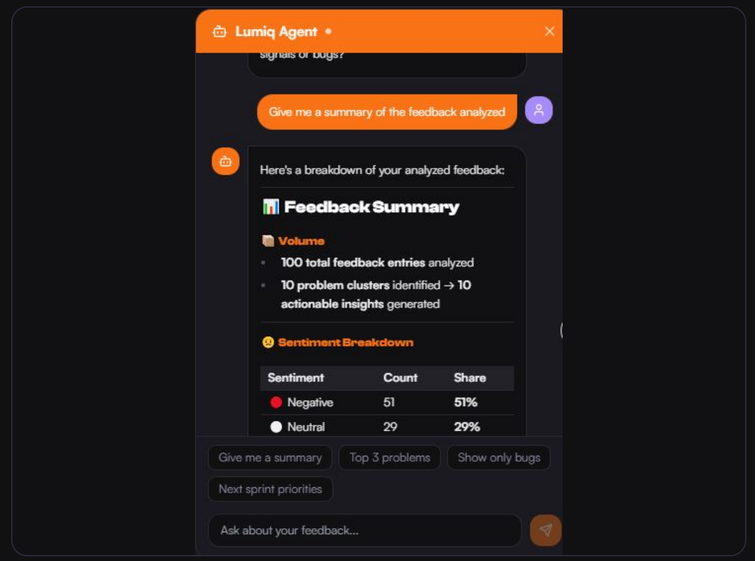
Dashboard

Prioritized insights with sentiment and intent breakdown



Cluster Explorer

Root cause per feedback theme with evidence trails



Conversational Agent

Ask questions about your data in plain English

View Live Product

Watch Walkthrough

RESULTS

What Success Looks Like

5 min

Average analysis time

For 100—200 feedback rows

~$0.09

Cost per analysis run

For 100 rows end to end

9

Autonomous pipeline steps

Zero human intervention required

Hours of manual feedback review **replaced with minutes** of AI analysis

Root causes identified — not just **surface-level complaint summaries**

Sprint-ready recommendations with **evidence trails PMs can trust**

Future metrics to track: Time saved per PM per week | Insights acted on per sprint | User satisfaction with recommendations

Lumiq

From feedback noise to product signal

Lumiq gives every product team the analytical power of a senior PM —
available instantly, at minimal cost.

Built by a PM
Who understands both the problem and the user

Portable to any company
GitHub hosted, self-deployable anywhere

Extensible by design
Every module independent and upgradeable

Lumiq — AI Product Feedback Agent • 0 to 1 PM Portfolio Project • 2026